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A comparative study of in-vitro and *in-silico* anti-candidal activity and GC–MS profiles of snow mountain garlic vs. normal garlic

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Abstract

Aim: The study aimed to profile the volatile phytocomposition of snow mountain garlic (SMG) compared to normal garlic and investigate the anti-*Candida* efficacy against clinically relevant multi-drug resistant isolates of *Candida* species.

Methods and Results: Herein, SMG has shown significantly superior fungicidal power at 2x-MIC dose against *C. albicans* and *C. glabrata* in killing kinetic evaluation unlike the fungistatic effect of normal garlic. GC–MS headspace-based profiling of SMG showed 5 unique volatile compounds and a 5-fold higher content of saponins than normal garlic. In an *in-silico* analysis, cholesta-4,6-dien-3-ol,(3-beta) was uniquely identified in SMG as a potential inhibitor with high binding affinity to the active site of exo-1,3-betaglucan synthase, an established anti-*candida* drug target crucial for the biofilm matrix formation, thus suggesting a plausible anti-*Candida* mechanism.

Conclusion: The in-vitro and *in-silico* studies have demonstrated the *Candida*-cidal and anti-biofilm activities of SMG, distinguishing it from the *Candida*-static efficacy of normal garlic.

Significance and Impact of the study: This is the first report that identifies several phytochemical signatures of SMG along with a potential anti-*Candida* compound, that is cholesta-4,6-dien-3-ol,(3-beta)-, which appears worthy of detailed studies in the future to explore the utility of SMG as a fungal phytotherapy agent, especially against drug-resistant *Candida* sp.

KEYWORDS

anti-fungal drug resistance, *Candida*-cidal activity, exo-1,3 beta glucan synthase, phytoactive compounds, snow mountain garlic

INTRODUCTION

An effective and safe anti-fungal treatment paradigm against a ubiquitous opportunistic fungal pathogen, *Candida*, remains a clinical challenge plagued with ever-increasing global morbidity, mortality and medical cost (Cortegiani et al., 2019). An upsurge in non-*albicans* Candidiasis (NAC), that is, caused by species other than *C. albicans* such as *C. glabrata*, *C. tropicalis*, *C. parapsilosis* and *C. auris* along with multi-drug resistant complications has become a grave cause of concern for clinical practitioners (Sanguinetti et al. 2015; Pristov and Ghannoum 2019). Among the cases of NAC, *C. tropicalis* (35.1%), *C. glabrata* (28.1%) and *C. krusei* (16.3%) are the top rankers showing resistance to first and second-generation (azole and echinocandins) anti-fungal drugs owing to long term usage of these synthetic anti-fungals (Deorukhkar et al. 2014a; Pristov and Ghannoum 2019). For example, according to Maubon and colleagues, *C. glabrata* has developed significant resistance to fluconazole (4–16%) and echinocandins (8–15%) (Maubon et al. 2014). As a result, the alarming rise in clinical candidiasis, combined with the limited efficacy of existing anti-fungal therapies, necessitates the search for newer, more effective and safe treatments. To improve therapeutic efficacy, these novel anti-fungals must be tested alone or in combination with existing therapies.

Phytotherapeutics is a rich natural resource worthy for bioprospecting novel anti-fungals. Moreover, owing to the general belief of plant-based interventions being safer, they have better patient compliance (Thamburan et al. 2006). Snow mountain garlic (SMG), locally named "*Kashmiri lahsun*" or "*Ek pothi lehsun*", is a wild *Allium* species that grows in the trans-Himalayan Jammu and Kashmir union territory of India. It is a rare single clove species of *Allium* that grows in subzero temperature at an elevation of 6000 feet above the mean sea level (Koul et al. 2016). Unlike common garlic or normal garlic (NG), each bulbous-shaped clove of SMG has an earthy brown hard shell, which naturally conserves its health benefits and also combats cold arid climate (Zúñiga-Martínez et al. 2019). The SMG bulbs, ranging between sizes of 1.5 to 2.5 cm, are peculiar in shape with stiff covering that is flat on one side and rounded on the other side, converging into a pointed tail or tip (Figure 1). Extremely localized growth in arid regions has limited its accessibility, and hence, scientific forays into its phytochemical, genetic and pharmacological profiles are nearly non-existent. However, practitioners of traditional medicine in the mountain states of India and Nepal, where SMG grows naturally and is consumed as a culinary additive, endorse its' anti-inflammatory, anti-arthritic and immunomodulatory properties (Shah 2014; Mahajan 2016). Moreover, NG is a well-established antimicrobial herb with several

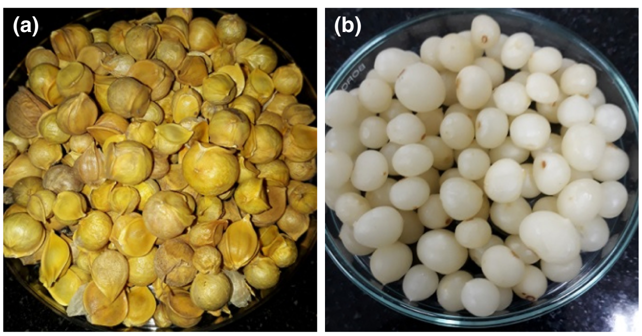

FIGURE 1 Picture of SMG. (a) unpeeled SMG cloves and (b) peeled SMG cloves

reports of its phytoconstituents responsible for the aforementioned activity against clinically relevant multi-drug resistant infective species. Nevertheless, to date, SMG has not been exploited scientifically for its potential anti-infective activity. Khodavandi and co-workers reported appreciable anti-*Candida* activity of NG targeting the hyphal cell wall protein (*HWP1*), an important structural constituent of the *Candida* biofilm (Khodavandi et al. 2011). Besides, NG has been recently shown to down-regulate the *Candida* virulence genes viz. *SIR1* (silent information regulatory gene) and *ECE1* (gene coding for candidalysin), which are associated with vulvovaginal *C. albicans* biofilm formation (Said et al. 2020).

Candida strains responsible for causing invasive candidiasis in immunocompromised patients are known to build an almost impermeable matrix using modified 1,3-beta-glucan substrate in their biofilms making them a thousand-fold more resistant than planktonic cells to most of the anti-fungals being used in clinics (Deorukhkar et al. 2014b). Interestingly, in the current *in-silico* molecular docking analysis, cholesta-4,6-dien-3beta-ol showed significantly strong binding affinity to the active site of exo-1,3 beta glucan synthase enzyme of *Candida*, which is an established drug target for fungal biofilm disruption. Authors propose cholesta-4,6-dien-3beta-ol as a potential anti-fungal phytocompound of SMG worth exploration for its anti-*Candida* efficacy and new drug development utility. The present study is a first exploration of the phytoconstituents of SMG, in comparison to the NG, accompanied by unprecedented bioactivity analysis of SMG in terms of anti-*Candida* activity against clinical *Candida* isolates.

The present study puts forth the profile of volatile phytoconstituents of SMG identified via the GC–MS-headspace technique and highlights its unique phytoconstituents in comparison to NG. Authors have investigated and described the anti-*Candida* efficacy of SMG extract in comparison to NG, evaluated by using drug susceptibility, MIC, killing kinetics, and biofilm eradication assays against clinical isolates of *Candida*, that is *C. albicans*, *C. glabrata* and *C. tropicalis*. Furthermore, the unique phytoactive compounds that differentiate SMG from NG were investigated with a special focus on phytocompounds with a high affinity for biofilms making target exo-1,3 beta glucan synthase.

MATERIALS AND METHODS

Sample collection, fungal species and chemical reagents

SMG bulbs were purchased from a local dealer from Jammu and Kashmir, India in June 2019, and NG (commercially cultivated) was collected from the local grower from Punjab (Dera Bassi village), India. The SMG sample was identified as *Allium ampeloprasum* L. (accession number: AC-55/2021) by Dr. Anjula Pandey, Division of Plant Exploration and Germplasm Collection, National Herbarium and Cultivated Plants, ICAR-NBPGR, New Delhi, India. Both species of garlic were stored at 4°C for the experimental studies. *C. albicans* MTCC 183, *C. tropicalis* MTCC 184 and *C. glabrata* MTCC 3019 strains were obtained from Microbial Type Culture Collection, CSIR-Institute of Microbial Technology (Chandigarh, India). Standard quality chemicals/reagents and media viz. vanillin (Sigma, USA), sulphuric acid (Rankem, India), yeast malt agar (Himedia, India), yeast malt broth (Himedia, India), amphotericin B (Bharat serum, India), fluconazole (Cipla, India), itraconazole (Himedia, India) and nystatin (Himedia, India) were purchased for the study.

Preparation of extracts

Bulbs of SMG and NG were extracted by conventional extraction method, that is, mechanical maceration, performed according to previously established protocol with few minor modifications (Lemar et al. 2002). Peeled bulbs (50 g) of both the garlic were homogenized with a hand blender in 100 ml of distilled water. The homogenized purees were vigorously stirred for 48 h at room temperature. Homogenates were filtered through 2–3 layers of cheesecloth to eliminate large solid particles followed by centrifugation at $6000 \times g$ for 20 min to separate small particles. The recovered filtrate was passed through sterile Whatman™ grade 1 filter paper. The aqueous extract of both types of garlic was freeze-dried and stored at −20 °C for further experimental use.

Qualitative and quantitative estimation of saponins

Qualitative and quantitative evaluation was done to estimate the total saponin content in an aqueous extract of SMG and NG.

Foam test (qualitative)

The foam test was performed according to the protocol described earlier (Mangathayaru 2013). To the 1 ml of extract of both the garlic was taken and further diluted 2-fold and then was shaken vigorously for 1 min. The development of stable foam in test tubes indicated the presence of saponins. Foam index was calculated by measuring the height of the stable froth thus formed. The formula for the foaming index is given as follows:

$$\text{Foaming index} = 1000 / \text{a},$$

where a = the volume in ml if the extract is used for preparing the dilution in the tube where foaming to a height of 1 cm is observed.

Emulsion test (qualitative)

One millilitre of olive oil was added to 2 ml of aqueous extracts of SMG and NG. The tubes were vortexed and observed for the formation of emulsion (Trease and Evans 1989).

Total saponin content (TSC) assay (quantitative)

The vanillin-sulphuric acid method was used to quantify the TSC, using a protocol previously established (Le et al. 2018). 0.25 ml of extract was incubated with 0.25 ml of vanillin (8% w v^{-1} in ethanol) and 2.5 ml of sulphuric acid (72% v v^{-1} in water). The reaction mixture was incubated at 60°C in a shaking incubator for 15 min. The reaction vials were cooled at room temperature for 5 min followed by the measurement of absorbance of the reaction mixture at 544 nm using a UV–VIS spectrophotometer (Molecular Devices, USA).

Gas chromatography–mass spectroscopy (GC–MS)

GC–MS-headspace analysis was performed to compare the volatile phytochemical composition of SMG vs. NG. The GC–MS analysis of each sample was carried out on Shimadzu (GC-2030) series GC–MS equipped with headspace (HS-20) (Shimadzu, Japan). The GC column SH-Rxi-5 SILMS (0.25 mm internal diameter × 30 m length × 0.25 μm film thickness) was used which was made of 1,4-bis(dimethylsiloxy)phenylene dimethylpolysiloxane. The flow rate of helium as carrier gas was set at 1.5 ml min^{-1}. The initial column temperature was set at 40°C for 5 min. The final temperature of the column was programmed to gradually raise to 250°C at the rate of 7°C held for 5 min, through split ratio (40:60) mode. Headspace condition was programmed as oven temp 80°C, sample line temp 150°C, transfer line temp 160°C, shaking level 4, multi-injection count 1, pressurizing

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gas pressure 50 kpa, equilibrating time 10 min, pressurizing time 1 min, injection time 1 min, needle flush time 1 min. The temperature of the ion source was set at 220°C and the interface temperature was 250°C. Samples were prepared by crushing freshly peeled bulbs of both the garlic, and 1 μl of the sample was injected at a constant temperature of 250°C through an autosampler injector. The ionization energy was 70 eV and the mass range of 40–500 AMU. The management of the GC–MS system, parameter settings for GC and mass spectrometry, and data receipt and processing were performed using Shimadzu real-time analysis. Identification of phytocompounds or their metabolites was performed via the NIST 2.0 (National Institute Standard and Technology) library database.

Susceptibility test (agar well diffusion assay)

Comparative preliminary anti-*Candida* susceptibility of aqueous extract SMG vs. NG was performed against three *Candida* sp. viz. *C. albicans*, *C. tropicalis* and *C. glabrata* using the agar well diffusion technique (Al Akeel et al. 2014). A single colony of each *Candida* sp. was inoculated in yeast malt broth (YMB) and activated overnight at 37°C. A 20 μl volume of the standardized inoculums (OD_{600} 0.4–0.6) was inoculated onto yeast malt agar (YMA) plates. Aqueous extracts of both the garlic sample adjusted to a stock concentration of 200 mg ml^{-1} were filtered through a sterile Whatman™ 0.22 μm PVDF membrane filter. Five concentrations of each extract were prepared i.e., 100, 80, 60, 40, 20 mg ml^{-1} from stock. 100 μl of each concentration of the aqueous extract was dispensed in a 6 mm well of agar plates and incubated at 37 °C for 24 and 48 h. Sterile water served as the negative control and 6 mm sterile paper disks soaked in standard drugs of concentration 100 μg disc^{-1} were placed on inoculated agar plates as positive control Agar plates were observed for the formation of a zone of inhibition (ZOI) after 24 and 48 h. The diameter of the ZOI in presence of anti-fungals was measured with a zone scale. The assay was performed in quintuplicate to estimate the reproducibility of the results. ZOI was measured which included the diameter of the well.

Minimum inhibitory concentration (MIC)

The MICs of the aqueous extracts of SMG and NG against clinical isolates of *Candida* sp. were determined by a 96-well microtiter plate dilution assay (Iscan et al., 2002). 20 μl of each concentration (0.8, 1.6, 3.2, 6.4, 12.8 and 25.6 mg ml^{-1}) of SMG and NG were dispensed in a 96-well plate. This was followed by the addition of 80 μl of YMB and 100 μl of activated cell suspension with a count of 2–5 × 10^{6} colony forming unit (CFU) per millilitre in the wells of the microtiter plate. Amphotericin B and fluconazole were used as standard drug control at concentrations 0.078 to 5 μg ml^{-1} and 25 to 400 μg ml^{-1}, respectively. Plate blank wells were poured with 200 μl of YMB and the positive control well contained 100 μl of fungal cell suspension and 100 μl of YMB. The plates were incubated at 37 °C and OD_{600} was measured at 24 and 48 h using an ELISA microplate reader (Thermo, USA).

Killing kinetics

Killing kinetics of SMG extract, NG extract, and amphotericin B was carried out against *Candida* sp. by a method described previously with slight modifications (Arunkumar et al. 2018). For time-kill studies, overnight activated culture of OD_{600} 1.2 to 1.4 was serially diluted to adjust initial CFU ml^{-1} approximately in the range of 2 × 10^{7} to 6 × 10^{7} [6.52 × 10^{7} (*C. albicans*), 4.86 × 10^{7} (*C. glabrata*), 2.66 × 10^{7} (*C. tropicalis*)]. Standardized cell suspension in YMB treated with MIC, 2x-MIC, 3x-MIC, and 4x-MIC of SMG, NG, and amphotericin B. Drug-free cell suspension served as an assay control. Incubation was done in shaking conditions at 37°C. For viability estimation, 0.1 ml of inoculums was withdrawn from each tube at predetermined time intervals (0, 2, 4, 6, 8, 10, 12, 24 and 48 h) and spread onto YMA plates to estimate the viable CFU. The number of colonies was counted after 24 h of incubation in standard conditions.

Biofilm eradication assay

Biofilm forming potential of clinical isolates of *Candida* sp. was evaluated in which 3 × 10^{5} to 8 × 10^{5} CFU ml^{-1} cells were incubated in YMB in 35 mm polystyrene petri plate for 24 and 48 h. Crystal violet staining was done for the microscopical examination of biofilm. The propensity of SMG extract to eradicate pre-formed biofilm was evaluated by a modified microtiter plate assay (Fernandes et al. 2020). Pre-formed biofilms were treated with the MIC, 2x-MIC and 3x-MIC of SMG, NG, and amphotericin B in a 96-well plate. Extract/drug-free biofilm was used as the positive control. After 24 h of incubation of the biofilm with test extracts, planktonic cells were removed and biofilm was re-suspended in phosphate buffer saline (pH 7.4). These re-suspended cells of biofilm were serially diluted to 10^{-4} to 10^{-5} CFU ml^{-1} and spread onto the YMA plate. Further, viable cells in the biofilm were estimated by the CFU count method.

Molecular docking of identified phytoconstituents from SMG against 1,3 beta glucan synthase

Docking simulation predicts the affinity of the ligands towards the active site of a biological target through the prediction of the interaction energy associated with each of the potential binding conformations (Ferreira et al., 2015). In the present study, exo-1,3 beta glucan synthase, which is a primary enzyme responsible for the synthesis of 1,3 beta glucan for building matrix of fungal biofilms, was selected for molecular docking analysis of volatile phytocompounds of SMG. The 3D crystal structure of exo-1,3 beta glucan synthase of *C. albicans* (PDB ID:1EQP), with 394 amino acid residues resolved to a resolution of 1.95 Å, was obtained from the Research Collaboratory for Structural Bioinformatics (RCSB) Protein Data Bank (https://www.rcsb.org/). The 3D spatial data files of six small molecules identified uniquely in SMG (Figure 6, Table S3) were downloaded from PubChem (https://pubchem.ncbi.nlm.nih.gov/). Structure-based docking simulations were performed using the AutoDock Vina tool compiled in UCSF Chimera 1.15 (Pettersen et al. 2004). The protein and the ligands were prepared using the integrated Dock Prep tool—the removal of water molecules, addition of hydrogens, assignment of Gasteiger charges to protein/ligand atoms and modelling of incomplete side chains (1EQP is a complete structure, no side chains were missing). The search space encompassed a grid with the following dimensions in Å: centre (x, y, z) = (31, 36, 56), dimensions (x, y, z) = (20, 20, 20). The docking simulation was then run at exhaustiveness of 8, the poses were returned based on the binding energy reported in kcal mol^{-1}. Docking protocol was optimized by re-docking the native ligand castanospermine to exo-1,3 beta glucan synthase (PDB ID: 1EQC) of *C. albicans* (Figure S5). Results and details of interaction were visualized using the BIOVIA Discovery Studio virtual environment and UCSF Chimera.

Statistical analysis

All the experiments were conducted in triplicate to establish the reproducibility of the results. Data were expressed in mean ± standard deviation. Statistical difference was analysed with multiple *t*-tests and three-way ANOVA followed by Turkey's post hoc test by IBM SPSS statistics 20. The p-value <0.05 was considered to be of statistical significance. In time kill studies $\log_{10}$ CFU ml^{-1} plotted against time to compare the rate of killing of standard drug and extracts.

RESULTS

Saponin content

The qualitative screening for saponins by using the foam test and emulsion test has indicated more enrichment of saponins in SMG than NG samples (Figure S4). In these tests, the SMG samples showed more stable foam and emulsion formation as compared to NG. The foaming index of SMG extract was found 5-fold higher than that of NG extract. For comparative quantification of the saponin content in SMG and NG samples, TSC assay was performed in which saponin content (includes steroidal sapogenins, triterpenoid sapogenins and sterol) was quantified and expressed as mg diosgenin equivalent (DE) g^{-1} of dry extract. The total saponin content of SMG was found significantly higher than that in NG with values of 90 ± 1.05 (9%) and 55.86 ± 0.83 (5.59%) mg DE g^{-1} respectively (p-value ≤0.005).

GC–MS-headspace based comparative phytochemical profile of SMG and NG

Comparative phytochemical components screening was done in freshly peeled garlic bulbs of SMG and NG (Tables S3 and S4**)**. Gas chromatography equipped with the headspace technique was used to fingerprint the active constituents in both the garlic. The mass spectra of phytocomponents identified in both the garlic samples were matched with the spectra of these compounds given the NIST 2.0 library database. Phytocompounds with their retention time, molecular formula, molecular weight, similarity index (SI) and percentage area have been provided. Enrichment of organosulfur compounds was identified in both the garlic. Interestingly, saturated fatty acid, and cholesterol derivative viz. pentadecanoic acid, and cholesta-4,6-dien-3-ol (3-beta) were identified uniquely in SMG and not in NG. Further, four organosulfur phytocomponents were identified exclusively in SMG bulbs, that is, methanethiol, dimethyl sulfide, propyl mercaptan and 1-allyl-2-isopropyldisulfane (Figure 6). Low concentration of organosulfur compounds such as methyl thiirane, allyl methyl sulfide, dimethyl disulfide, diallyl sulfide, methyl 2-propenyl disulfide, methyl propyl disulfide, diallyl disulfide, (E)-1-allyl-2-(prop-1-en-1-yl) disulfane, methyl 2-propenyl trisulfide, 2-vinyl-4H-1,3-dithiine, and di-2-propenyl trisulfide in SMG could explain its mild alliaceous odour and flavour, unlike NG which was found to have a high concentration of volatile organosulfur compounds and thus a strong alliaceous odour and flavour too.

Susceptibility to SMG and NG

The susceptibility of all three clinical isolates of *Candida* sp. was tested against standard anti-fungal drugs used in clinics. Nystatin showed maximum efficacy with ZOI of 20.33 ± 0.58, 17.67 ± 0.58 and 20.33 ± 0.58 mm against *C. albicans*, *C. tropicalis*, and *C. glabrata* respectively. All the tested *Candida* isolates were found partially resistant to fluconazole and itraconazole. (Table S1 and Figure S1).

SMG extract exhibited a significant inhibitory effect against all the three tested *Candida* sp. viz. *C. albicans*, *C. tropicalis*, and *C. glabrata*. Fungal species tested in the present study were susceptible to all six concentrations of SMG, i.e., 20, 40, 60, 80, 100, 200 mg ml^{-1} (Tables 1–3). Dose and time-dependent inhibitory effects of SMG were compared with NG's anti-*Candida* activity (Figure S2). The study indicated significantly superior anti-*Candida* efficacy of SMG than NG. The ZOI observed at the lowest tested dose of SMG i.e., 20 mg ml^{-1} was 10.8 ± 1.30, 11.2 ± 1.10, and 10.4 ± 0.55 mm against *C. albicans*, *C. tropicalis*, and *C. glabrata* at 24 h, respectively (Table 1–3). Whereas minimum effective doses of NG were 40, 80, and 60 mg ml^{-1} with ZOI of 10 ± 0, 10.2 ± 0.45, and 10.8 ± 1.30 mm against *C. albicans*, *C. tropicalis*, and *C. glabrata* at 24 h, respectively. Interestingly, the inhibition zones on the SMG-treated agar plates were found intact up to 48 h in almost all the tested concentrations whereas this was not the case with ZOIs developed by NG treatment. At 48 h duration, the minimum effective doses of SMG and NG were 40 mg ml^{-1} and 200 mg ml^{-1} against all three *Candida* sp., respectively. However, with prolonged incubation, a re-growth in the ZOIs indicated only the fungistatic efficacy of NG whereas sustained ZOIs in SMG indicated a strong fungicidal activity in SMG.

TABLE 1 Zone of inhibition (mm) against *C. albicans*

Conc. ($mg\,ml^{-1}$)	24 h SMG	24 h NG	48 h SMG	48 h NG
20	10.8 ± 1.30	0 ± 0 $^{\Psi}$	0 ± 0$^{\delta}$	0 ± 0
40	13.6 ± 0.55	10 ± 0 $^{\Psi}$	11.8 ± 1.30	0 ± 0 $^{\Psi\,@}$
60	14.8 ± 1.10	12 ± 1.22 $^{\Psi}$	13.6 ± 1.14	0 ± 0$^{\Psi\,@}$
80	16.4 ± 1.34	13.8 ± 0.84$^{\$}$	14.4 ± 0.89	0 ± 0$^{\Psi\,@}$
100	17.6 ± 1.52	17.2 ± 1.64	16.2 ± 0.84	10.8 ± 0.84 $^{\Psi\,@}$
200	19.2 ± 1.30	20.4 ± 1.14	18.4 ± 0.55	16 ± 1.58$^{\$\,@}$

Note: Data expressed as mean ± SD (*N* = 5). Statistical difference was calculated by using multiple *t*-test. Statistical significance difference regime: SMG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.05$^{\$}$ and ≤0.001$^{\Psi}$; SMG vs. SMG at the time interval of 24 and 48 h *p*-value ≤0.001$^{\delta}$; NG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.001$^{@}$.

Abbreviations: NG, normal garlic; SMG, snow mountain garlic.

TABLE 2 Zone of inhibition (mm) against *C. tropicalis*

Conc. ($mg\,ml^{-1}$)	24 h SMG	24 h NG	48 h SMG	48 h NG
20	11.2 ± 1.10	0 ± 0$^{\Psi}$	0 ± 0$^{\delta}$	0 ± 0
40	13.2 ± 1.10	0 ± 0$^{\Psi}$	12 ± 1	0 ± 0$^{\Psi}$
60	14.6 ± 1.34	0 ± 0$^{\Psi}$	13.4 ± 1.52	0 ± 0$^{\Psi}$
80	15.4 ± 1.52	10.2 ± 0.45$^{\Psi}$	14.4 ± 1.52	0 ± 0$^{\Psi\,@}$
100	16.4 ± 1.52	11.6 ± 0.55$^{\Psi}$	15 ± 1.22	0 ± 0$^{\Psi\,@}$
200	19 ± 0.71	17.2 ± 1.10$^{\$}$	18 ± 1.41	12.2 ± 1.30$^{\Psi\,@}$

Note: Data expressed as mean ± SD (N = 5). Statistical difference was calculated by using multiple *t*-test. Statistical significance difference regime: SMG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.05$^{\$}$ and ≤0.001$^{\Psi}$; SMG vs. SMG at the time interval of 24and 48 h *p*-value ≤0.001$^{\delta}$; NG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.001$^{@}$.

Abbreviations: NG, normal garlic; SMG, snow mountain garlic.

TABLE 3 Zone of inhibition (mm) against *C. glabrata*

Conc. ($mg\,ml^{-1}$)	24 h SMG	24 h NG	48 h SMG	48 h NG
20	10.4 ± 0.55	0 ± 0$^{\Psi}$	0 ± 0$^{\delta}$	0 ± 0
40	12.2 ± 1.10	0 ± 0$^{\Psi}$	11 ± 1.22	0 ± 0$^{\Psi}$
60	13.8 ± 0.84	10.8 ± 1.30$^{\#}$	12.4 ± 1.14	0 ± 0$^{\Psi\,@}$
80	15.6 ± 0.55	13.2 ± 2.17	13.8 ± 1.30	0 ± 0$^{\Psi\,@}$
100	16.4 ± 0.55	17.4 ± 1.14	15.2 ± 1.10	0 ± 0$^{\Psi\,@}$
200	19.8 ± 0.45	20.4 ± 1.14	18.8 ± 0.84	12.4 ± 1.14$^{\Psi\,@}$

Note: Data expressed as mean ± SD (*N* = 5). Statistical difference was calculated by using multiple *t*-test. Statistical significance difference regime: SMG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.05$^{\$}$, ≤0.005$^{\#}$, and ≤0.001$^{\Psi}$; SMG vs. SMG at the time interval of 24 and 48 h *p*-value ≤0.001$^{\delta}$; NG vs. NG at the time interval of 24 and 48 h *p*-value ≤0.001$^{@}$.

Abbreviations: NG, normal garlic; SMG, snow mountain garlic.

Minimum inhibitory concentration (MIC)

In-vitro MIC of aqueous extract of SMG and NG against clinical isolates of *Candida* sp. was evaluated for 24 and 48 h time durations (Table S2). All three *Candida* sp. showed distinct susceptibility patterns to both the garlic extracts. *C. tropicalis* has shown higher susceptibility to SMG with MIC of 1.6 mg ml^{-1} which was 8-fold lower than the MIC of NG which was 12.8 mg ml^{-1}. Of all the *Candida* sp., *C. glabrata* was the most resistant strain that showed MIC of 6.4 mg ml^{-1} and 12.8 mg ml^{-1} for SMG and NG, respectively. At 24 h, the susceptibility pattern of *Candida* isolates for SMG was *C. tropicalis* > *C. albicans* > *C. glabrata* while at 48 h it was *C. tropicalis*≈ *C. albicans* > *C. glabrata*. Most importantly, the anti-*Candida* efficacy of SMG was sustained up to 48 h in the case of *C. albicans* and *C. glabrata* whereas a two-fold increase in MIC of NG was observed against all three *Candida* sp. at 48 h. The susceptibility of

Candida isolates for NG was the same at 24 and 48 h i.e., *C. albicans* > *C. tropicalis* ≈ *C. glabrata*. According to Clinical and Laboratory Standards Institute (CLSI), approved standard M27-A3, *Candida* sp. with cut-off MIC >1 μg ml^{-1} were considered as resistant; hence, *C. tropicalis* was considered to be resistant to amphotericin B with MIC 1.25 μg ml^{-1} whereas resistance to fluconazole was observed in all the tested isolates (CLSI M27-A3 2008).

Killing kinetics of SMG and NG

For evaluation, the fungicidal effect was considered when the reduction in the number of CFU ml^{-1} was greater than 3 log$_{10}$ units (99.9%) and fungistatic effect as <99.9% (<3 log$_{10}$) reduction in CFU ml^{-1} from the starting inoculums (Cantón et al. 2004). The aqueous extracts of SMG and NG were evaluated and compared for respective fungicidal and fungistatic effects at MIC, 2x-MIC, 3x-MIC and 4x-MIC. In the case of *C. albicans* SMG exhibited fungicidal effect (>3 log$_{10}$ CFU ml^{-1}, 99.9%) at a concentration of 2x-MIC at 2 h time point whereas at MIC dose this garlic showed fungistatic activity (<3 log$_{10}$ CFU ml^{-1}), which was significantly higher than that of NG and amphotericin B (Figure 2). In the case of *C. glabrata*, SMG extract showed fungicidal effect at 3x and 4x-MICs at 2 h time point and fungistatic activity with MIC dose. Whereas at 4 h time point even 2x MIC of SMG could exhibit a fungicidal effect. In comparison to this, all tested doses of NG were found to be fungistatic only (<3 log$_{10}$ CFU ml^{-1}). The standard drug, amphotericin B could also show fungicidal effect but only at 4x-MIC dose, and all other doses i.e., MIC, 2x-MIC, and 3x-MIC were found fungistatic (Figure 3). In the case of *C. tropicalis*, only 4x-MIC of SMG could show fungicidal activity, whereas all other doses were only fungistatic. In this study, NG and amphotericin B were found only fungistatic for *C. tropicalis* at the highest tested concentration (Figure 4).

Biofilm eradication potential of SMG

The biofilm-forming potential of all three *Candida* isolates was evaluated microscopically. *C. tropicalis* showed the strongest multilayer biofilm formation with intense hyphae and pseudohyphae formation whereas *C. albicans* showed moderate potential to form a biofilm (Figure S3). In the present study, *C. glabrata* failed to form stable biofilms, and thus was not studied further. Biofilm eradication potentials of SMG, NG and amphotericin B were thus estimated against *C. albicans* and *C. tropicalis* only. SMG could significantly reduce the biofilm burden of *C. albicans* in comparison to amphotericin B and NG with the lowest CFU at MIC, 2x-MIC & 3x-MIC, that is, 11.67 ± 0.17, 11.13 ± 0.14 & 9.26 ± 0.13 CFU cm^{-2}, respectively (Table 4). All the tested MICs of SMG were also found equally effective against *C. albicans* (Figure 5a). Hence, biofilm eradication efficacies of two garlic extracts and drug control for *C. albicans* were in the order of SMG > amphotericin B > NG. However, the potency of SMG extract to reduce the burden of viable cells in the biofilm of *C. tropicalis* was found at par with amphotericin B yet significantly superior to NG (Figure 5b). The CFU values at MIC and 3x-MIC of SMG extract were 4.37 ± 0.12 and 2.97 ± 0.10 CFU cm^{-2}, whereas for amphotericin B,

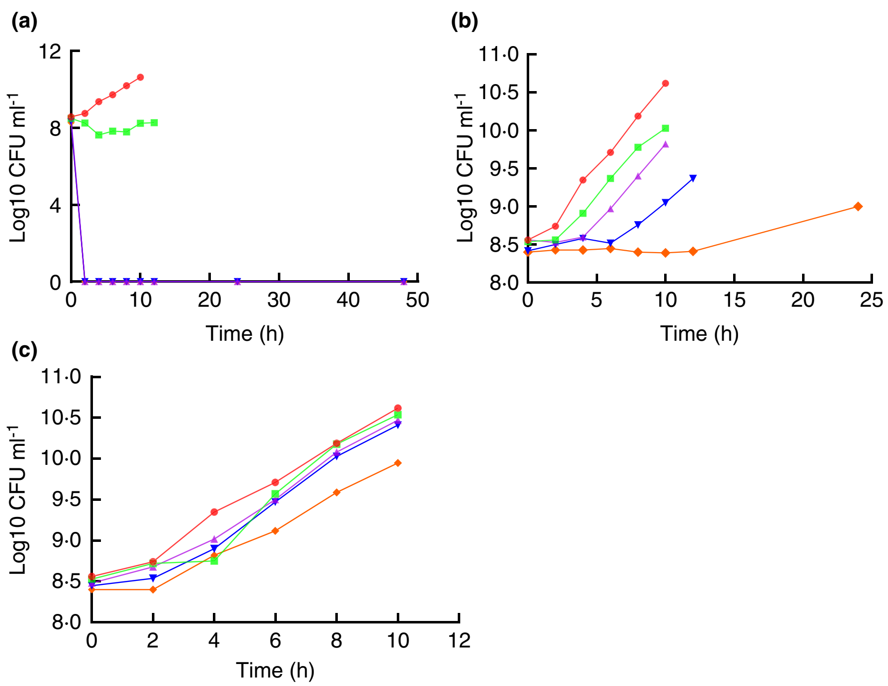

FIGURE 2 Time kill kinetics against *C. albicans* by SMG (a), NG (b), and amphotericin B (c) at different concentrations: Control (●), MIC (■), 2x-MIC (▲), 3x-MIC (▼) and 4x-MIC (◆) were tested for fungicidal effect at several time points (x-axis). CFU count method was used to measure the viability of the treated cells.

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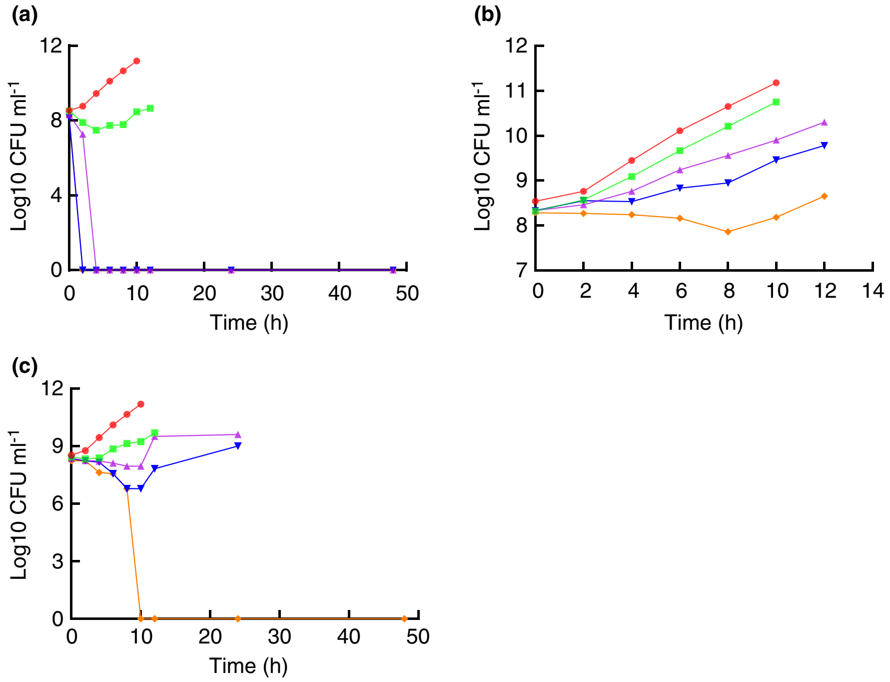

FIGURE 3 Time kill kinetics against *C. glabrata* by SMG (a), NG (b) and amphotericin B (c) at different concentrations: Control (●), MIC (■), 2x-MIC (▲), 3x-MIC (▼) and 4x-MIC (♦) were tested for fungicidal effect at several time points (x-axis). CFU count method was used to measure the viability of the treated cells.

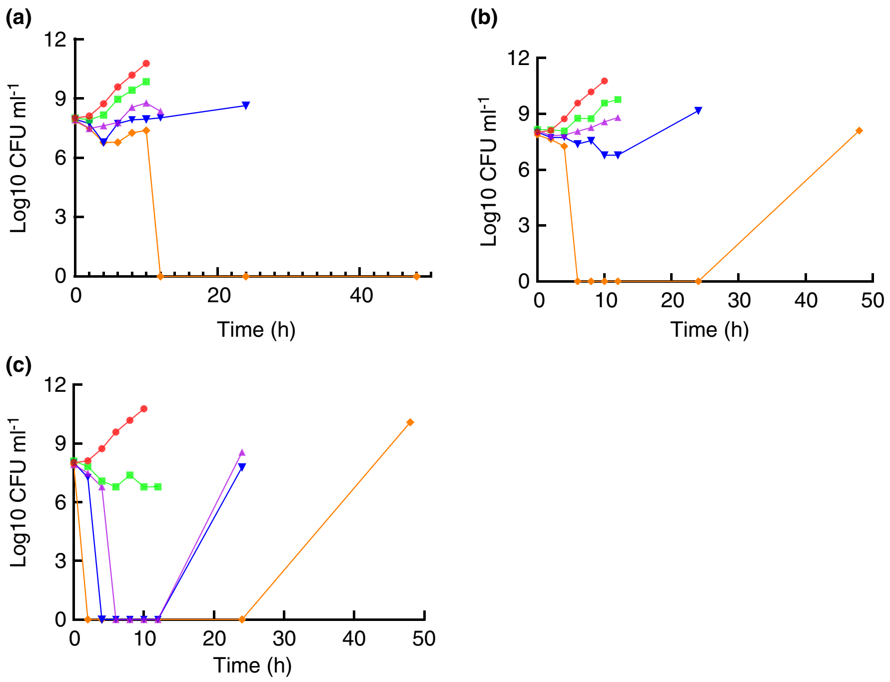

FIGURE 4 Time kill kinetics against *C. tropicalis* by SMG (a), NG (b) and amphotericin B (c) at different concentrations: Control (●), MIC (■), 2x-MIC (▲), 3x-MIC (▼) and 4x-MIC (♦) were tested for fungicidal effect at several time points (x-axis). CFU count method was used to measure the viability of the treated cells.

TABLE 4 Viable cells of *C. albicans* biofilm after extract/drug treatment for 24 h

	CFU $cm^{-2}\times10^{5}$			
Conc. ($mg\,ml^{-1}$)	**SMG**	**NG**	**AMB**	**Control**
MIC	$11.67\pm0.29^{*\Psi\#}$	$60.18\pm0.54^{*}$	$45.46\pm0.76^{*@}$	64.21 ± 0.59
2x-MIC	$11.13\pm0.25^{*\Psi\#}$	$59.51\pm0.52^{*}$	$29.26\pm0.68^{*@}$	64.21 ± 0.59
3x-MIC	$9.26\pm0.23^{*\Psi\#}$	$57.78\pm0.63^{*}$	$22.76\pm0.69^{*@}$	64.21 ± 0.59

Note: The values are expressed as mean ± SD of a triplicate set. Statistical significance difference was calculated by three-way ANOVA followed by Turkey's post hoc test. Statistical significance difference regime at *p*-value ≤0.001: control vs. SMG/NG/AMB*; SMG vs. NG$^{\Psi}$; SMG vs. AMB$^{\#}$; and AMB vs. NG$^{@}$.
Abbreviations: AMB, amphotericin B; NG, normal garlic; SMG: snow mountain garlic.

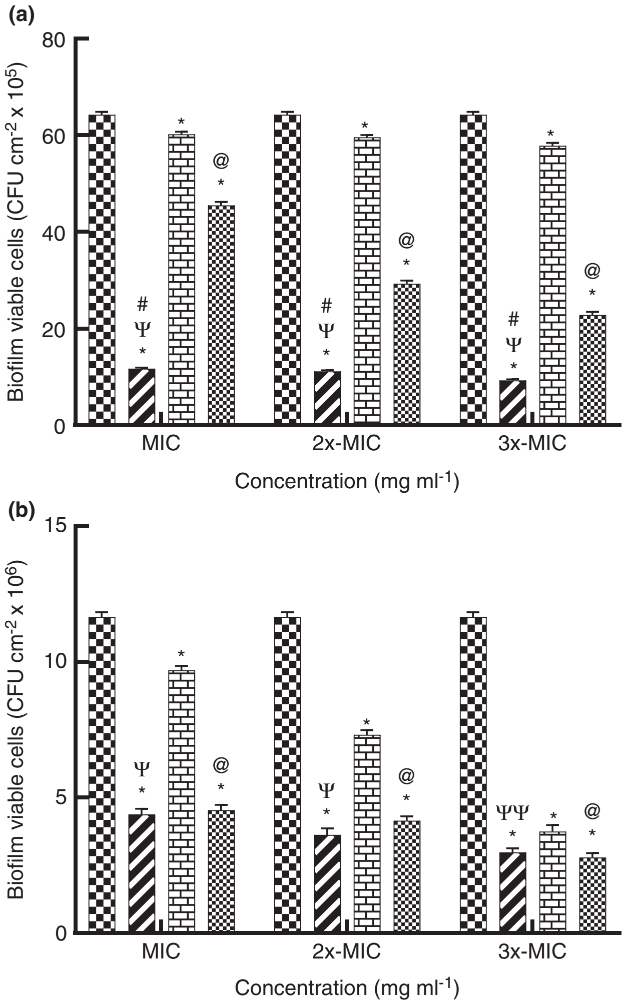

FIGURE 5 Viable cell estimation of *C. albicans* (a) and *C. tropicalis* (b) biofilm treated with MIC, 2x-MIC and 3x-MIC of () SMG, () NG,() amphotericin B, and () control. Biofilm eradication potential of extract/drug was determined by CFU count method (CFU cm^{-2} surface area of petri plate). Statistical significance difference was calculated by three-way ANOVA followed by Turkey's post hoc test. Statistical significance difference regime at *p*-value ≤0.001: Control vs. SMG/NG/AMB*; SMG vs. NG^{Ψ}; SMG vs. $AMB^{\#}$; and AMB vs. $NG^{@}$. Greek letter in pair ($^{\Psi\ \Psi}$) show significance at *p*-value ≤0.005.

it was 4.53 ± 0.12 and 2.78 ± 0.10 CFU cm^{-2}, respectively (Table 5). Thus, biofilm reduction efficacy for *C. tropicalis* was in the order of SMG≥ amphotericin B > NG.

In-silico molecular docking of identified phytoconstituents from SMG against 1,3 beta glucan synthase from *Candida*

The RMSD value between the atomic coordinates of co-crystallized castanospermine retrieved from PDB and the top docked pose was 0.3 Å, hence the same docking protocol was further implemented. Six molecules unique to SMG were assessed for their binding potential to the active site of exo-1,3 beta glucan synthase (1EQP) and are reported in Table 6, ranked with their respective binding energies (kcal mol^{-1}). Low binding free energy indicates a stable and favoured protein-ligand complex. CID 14795191,that is, cholesta-4,6-dien-3-ol binding with the active site of exo-1,3 beta glucan synthase had the lowest binding free energy (−9.7 kcal mol^{-1}). The hydroxyl residue of cholesta-4,6-dien-3-ol inserts in the keyhole-shaped active site of the enzyme and forms non-covalent bonds with the key residues of the active site—hydrogen bond with Glu27 and Trp363 and hydrophobic interactions with Trp 373, Tyr 255, Phe144, Leu194, Phe229, and Phe258, as seen in Figure 7a (also refer Figure S6). The binding energy for the remaining five molecules in SMG was high, and the molecules are unlikely to bind the active site of exo-1,3 beta glucan synthase.

DISCUSSION

Members of *Candida* sp., despite being a human commensal, are increasingly causing local and systemic infections, with moderate to severe impact on the infected subject's health. *C. albicans*, along with NAC (*C. glabrata*, *C. tropicalis*, *C. parapsilosis*, *C. dubliniensis*, *C. guilliermondii*, *C.*

TABLE 5 Viable cells of *C. tropicalis* biofilm after extract/drug treatment for 24 h

	CFU $cm^{-2} \times 10^6$			
Conc. (mg ml^{-1})	**SMG**	**NG**	**AMB**	**Control**
MIC	4.37 ± 0.21* $^{\Psi}$	9.68 ± 0.17*	4.53 ± 0.20*$^{@}$	11.65 ± 0.18
2x-MIC	3.61 ± 0.25* $^{\Psi}$	7.30 ± 0.18*	4.14 ± 0.16*$^{@}$	11.65 ± 0.18
3x-MIC	2.97 ± 0.17* $^{\Psi\Psi}$	3.74 ± 0.23*	2.78 ± 0.17*$^{@}$	11.65 ± 0.18

Note: The values are expressed as mean ± SD of a triplicate set. Statistical significance difference was calculated by three-way ANOVA followed by Turkey's post hoc test. Statistical significance difference regime at *p*-value ≤0.001: control vs. SMG/NG/AMB*; SMG vs. NG $^{\Psi}$; SMG vs. $AMB^{\#}$; and AMB vs. NG $^{@}$. Greek letters in pair ($^{\Psi\Psi}$) show significance at *p*-value ≤0.005.

Abbreviations: AMB, amphotericin B; NG, normal garlic; SMG: snow mountain garlic.

TABLE 6 Binding affinity of uniquely identified molecules in SMG to the active site of exo-1,3 beta glucan synthase crystal structure (PDB ID: 1EQP) determined using molecular docking

S.No.	PubChem ID/Name	Binding Affinity (kcal mol^{-1}) with 1EQP active site
1	CID 14795191/ Cholesta-4,6-dien-3-ol, (3-beta)	−9.7
2	CID 13849/ Pentadecanoic acid	−6
6	CID 525503/1-Allyl-2-isopropyldisulfane	−3.8
5	CID 7848/ Propyl mercaptan	−2.6
3	CID 12232/Dimethyl disulfide	−2.1
4	CID 878/Methanethiol	−1.2
7	Castanospermine (redocked native ligand of 1EQC, CID 54445)	−7.8

krusei and *C. kefyr*) are the third most dominant cause of nosocomial infections. The expanding resistance mechanisms to known anti-fungal interventions add to the medical severity of candidiasis. Further, mild to severe toxic side effects of anti-fungal drugs are a cause for concern, due to the evolutionary nearness of fungi to a human host, as compared to prokaryotic infectious microbes (Martins et al. 2015; Seneviratne and Rosa 2016; Taei et al. 2019). Phytomolecules have a long-standing history of being an important source for new drug discovery—between 1981 to 2014 approximately half the novel drugs approved by the United States Food and Drug Administration were either sourced naturally or developed from a natural molecule (Newman and Cragg 2012). Despite their immense therapeutic potentials along with much lower safety concerns, the bioactive phytocompounds remain an underexplored resource for safer and more effective anti-fungal agents to date. Garlic (*A. sativum*) is a widely used culinary additive, with established anti-microbial properties (against prokaryotic and eukaryotic infectious agents), owing to its phytochemical constituents (Nakamoto et al. 2020, Perry et al. 2009). In the present study, SMG, a trans-Himalayan species of garlic, is profiled for its unique phytoconstituent composition and is further evaluated via in-vitro and *in-silico* anti-*Candida* potential against the *C. albicans*, and most prevalent clinically relevant NAC viz *C. tropicalis* and *C. glabrata*.

SMG is a perennial tuberous plant that can naturally survive temperatures as low as −10°C during peak winters (Koul et al. 2016). As mentioned earlier, being a regional species with limited geographical access, scientific explorations on the phytochemical constitution or bioactivity-based characterizations of SMG have remained scanty to date. In this study, the authors provide the first report on phytocompositional analysis of SMG, we observed a manifold high content of steroidal saponins, as well as certain unique volatile bioactive molecules. Saponins are considered potent anti-fungal phytocompounds and have been extracted from various species of genus *Allium* (Lanzotti et al. 2014; Sobolewska et al. 2016). Morita and co-workers reported potent anti-fungal activity of another regional species of garlic—elephant garlic—against *Aspergillus niger* and *C. albicans* owing to its high steroidal content (Morita et al. 1988). Steroidal saponins are predicted to have binding affinity to the cell membrane localized ergosterol causing pore formation in the membrane and consequently leading to osmotic stress and cell death i.e., fungicidal effect (Simons et al. 2006; Cho et al. 2013).

Besides higher content of saponins and other popularly studied organosulphur compounds of *Allium* plants (e.g. allicin, ajoene and diallyl trisulfide), it is being proposed that uniquely identified phytocompounds of SMG in the present study viz four organosulphur compounds methanethiol, dimethyl sulfide, propyl mercaptan and 1-allyl-2-isopropyldisulfane, along with pentadecanoic acid, and cholesta-4,6-dien-3-ol (Figure 6; Table S3), could be collectively responsible for the significantly higher anti-*Candida* activity of SMG extract, in comparison to the extract of NG as observed in the present study (Lewis 1985; Yoshida et al. 1987; Shadkchan et al. 2004; Davis 2005; Hayat et al. 2016; Zhu et al. 2018; Wang et al. 2019; Bhattacharyya et al. 2020; Toral et al. 2021).

In an attempt to further investigate the anti-*Candida* bioactivity of SMG and ascribe a potential molecule to the observed superior fungicidal efficacy of SMG in comparison to NG, exo-1,3 beta glucan synthase was chosen as the protein target for in silico molecular docking and binding analysis. Exo-1,3 beta glucan synthase is a membrane-associated protein responsible for the synthesis of β-1,3-d-glucan polymers composing the fungal cell wall and eventually drug-resistant matrix of fungal biofilm responsible for the severity of candidiasis. Exo-1,3 beta glucan synthase is the target of a clinically successful class of anti-fungal drugs, viz., echinocandin, pneumocandins and ibrexafungin, which target an allosteric site away from the active site of the enzyme. Inhibition of glucan synthases is a safer and less toxic pharmacological strategy than the inhibition of lanosterol 14α-demethylase by azole drugs (Walker et al. 2011; Lima et al. 2019). The activity of exo-1,3 beta glucan synthase relies on the recognition and stabilization of sugar residue, in the extending glucan chain, via a hydrogen bond with the C3 and C4 hydroxyl of the sugar with the carboxyl groups in the Glu27, so that the C1 position is accessible by the nucleophile Glu292 and proton donor Glu192. Besides, there is a hydrophobic clamp at the entry to the active site formed by the flanking Phe144 and Phe 258 which holds the incoming

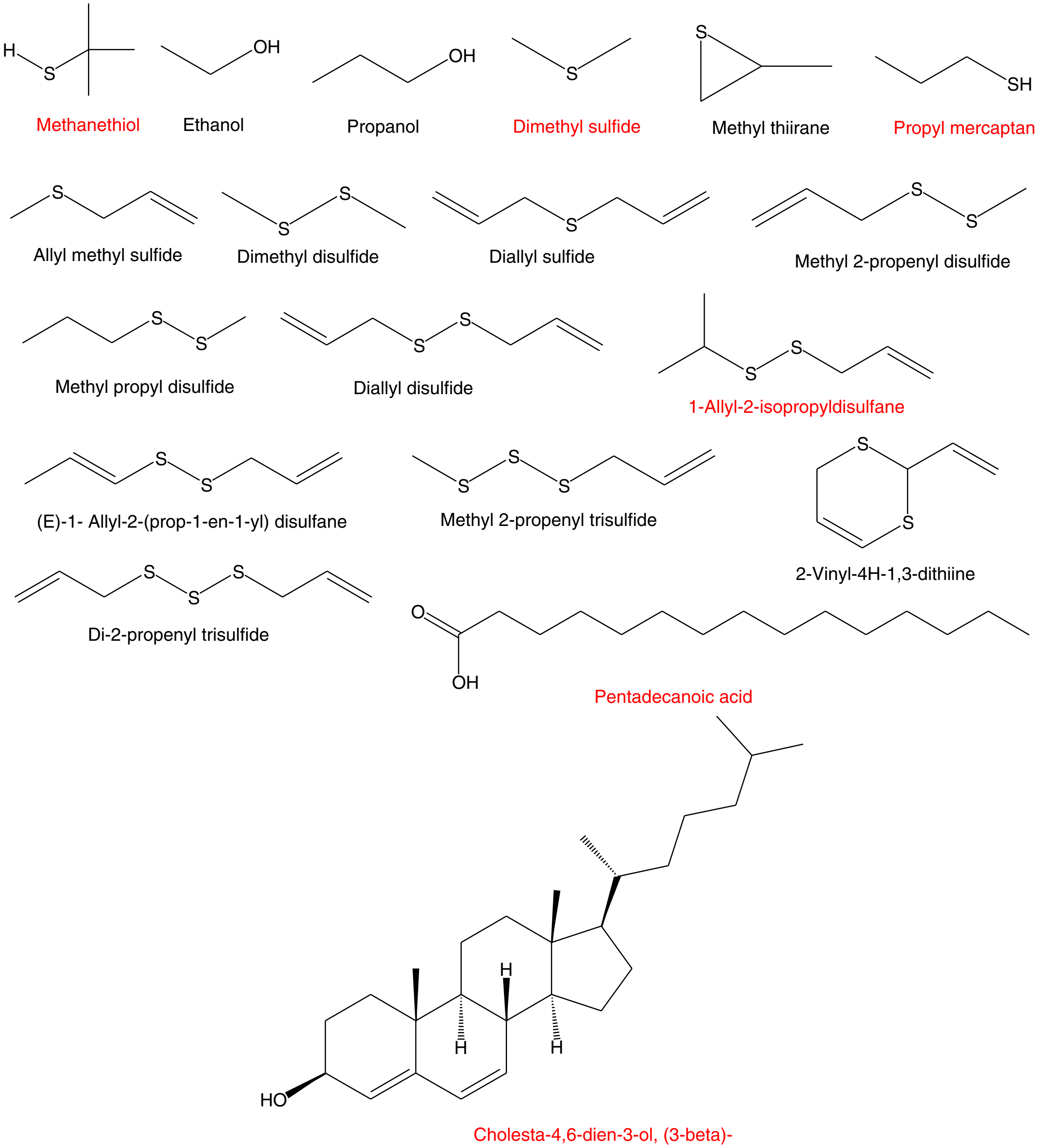

FIGURE 6 Phytocompounds identified in SMG bulbs. Phytocompounds names in red colour bold font style were uniquely identified in SMG samples and were not found in NG samples.

sugar residue facilitating the extension of the glucan chain (Cutfield et al. 1999). Our molecular docking analyses demonstrate that cholesta-4,6-dien-3-ol, binds the active site of the exo-1,3 beta glucan synthase by blocking the key residues—Glu27, Phe144 and Phe 258— besides interacting with other residues in the active site, forming a stable and favourable binding complex (Figure 7b, Table 6). Cholesta-4,6-dien-3-ol, isolated from marine worms, has previously demonstrated potent antimicrobial activity against select bacteria and fungi (Zhu et al. 2018). Our *in-silico* investigation is the first report of cholesta-4,6-dien-3-ol as a potential anti-*Candida* moiety targeting the active site of glucan synthases in *Candida* sp.

It is imperative here to emphasize that the superior anti-*Candida* activity of SMG in comparison to NG, established via susceptibility studies, MIC determination, time-kill kinetics, and biofilm disruption assays implicates a more fungicidal nature as compared to fungistatic action. For example, in susceptibility studies, the ZOI produced by a manifold lower dose of SMG aqueous extract

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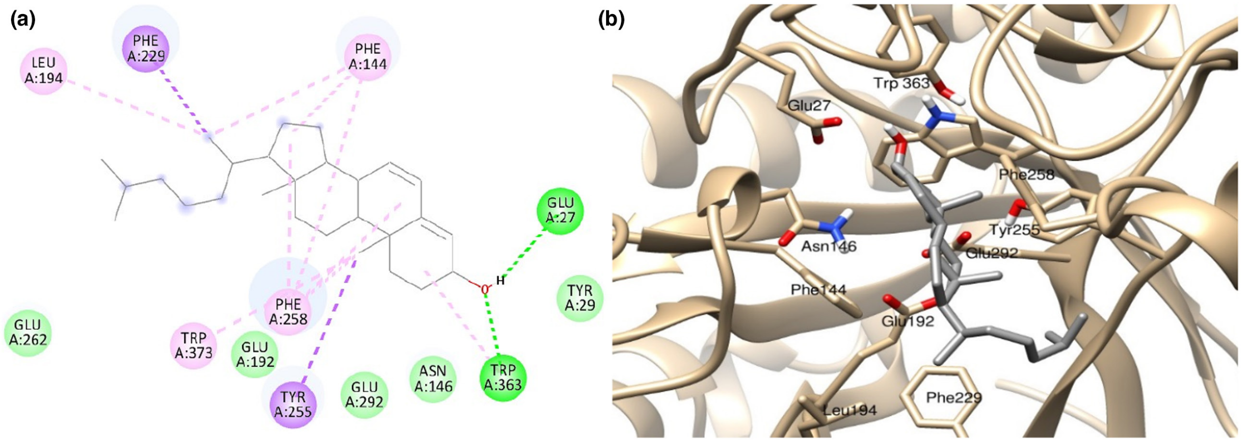

FIGURE 7 2D ligand interaction map of CID14795191 with key amino acid residues on the 1EQP active site (a). The green dotted lines depict hydrogen bonds, purple dotted lines depict pi-sigma hydrophobic interaction, and pink dotted lines depict pi-alkyl and pi-pi hydrophobic interactions, contributing to stability and favourable binding of the molecule to the active site. Figure (b) represents CID14795191 top docked pose (grey) in the vicinity of the key amino acid residues on the 1EQP active site.

was maintained up to 48 h time interval, against all three clinically relevant *Candida* sp. tested (Figure S2) indicating a fungicidal action, against the much higher dose of aqueous extract of NG which could maintain the ZOI only up to 24 h thus indicating fungistatic action. Assuming the inhibition of exo-1,3 beta glucan synthase by cholesta-4,6-dien-3-ol in SMG as a key anti-fungal mechanism, the fungicidal mode of action of SMG can be proposed, since it is well established that inhibition of glucan synthases by echinocandins is fungicidal in the *Candida* genus (Lewis and Graybill, 2008; Sucher et al. 2009).

There have been an increase in drug-resistant strains of *Candida* infections (*albicans* and NAC), wherein higher MIC values are emerging against azole drugs as compared to echinocandins, albeit echinocandin resistance is also on the rise (Pristov and Ghannoum, 2019). Interestingly, all three *Candida* sp. implied in the present study were found resistant to the azole class of anti-fungals (i.e., fluconazole and itraconazole), but were susceptible to SMG/ NG extracts, notwithstanding echinocandin class of drugs as a positive control was not included in our experimentation.

Summarily, the present study is the first-ever report generated on the distinctive phytochemical profile and anti-*Candida* efficacy of SMG, in comparison to NG. Given the present observations, SMG could offer a potential solution to the various facets of medical hurdles associated with *Candida* infection—safer and better compliance by the end-user due to its plant origin. The fungicidal action of SMG may improve the therapeutic outcomes in infected subjects, owing to its plausible inhibition of fungal glucan synthases, a safer drug target than azole mediated lanosterol 14α-demethylase inhibition, thus overriding the widening drug resistance mechanisms in *Candida* sp. due to its unique phytoconstituents with strong potential for being explored as anti-fungal lead molecules (saponins, cholesta-4,6-dien-3-ol). However, further in-depth clinical investigation on the proposed as well as other possible modes of action of unique phytoactive components from SMG is recommended.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest concerning the publication of this paper.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are available from the corresponding author upon request.

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REFERENCES

Al Akeel, R., Al-Sheikh, Y., Mateen, A., Syed, R., Janardhan, K. & Gupta, V.C. (2014) Evaluation of antibacterial activity of crude protein extracts from seeds of six different medical plants against standard bacterial strains. *Saudi Journal of Biological Sciences*, 21, 147–151. https://doi.org/10.1016/j.sjbs.2013.09.003

Arunkumar, K., Ehsanollah, G.R., Jayakayatri, N.J., Fazlin, M.F., Leslie, T.L.T., Mallikarjuna, P.R. et al. (2018) *Anti-Candida* and antibiofilm activity of Persian shallot (*Allium stipitatum* regel.) against clinically significant *Candida* spp. *Tropical Biomedicine*, 35, 815–825.

Bhattacharyya, A., Sinha, M., Singh, H., Patel, R.S., Ghosh, S., Sardana, K. et al. (2020) Mechanistic insight into the antifungal effects of a fatty acid derivative against drug-resistant fungal infections. *Frontiers in Microbiology*, 11, 2116. https://doi.org/10.3389/fmicb.2020.02116

Cantón, E., Pemán, J., Gobernado, M., Viudes, A. & Espinel-Ingroff, A. (2004) Patterns of amphotericin B killing kinetics against seven *Candida* species. *Antimicrobial Agents and Chemotherapy*, 48, 2477–2482. https://doi.org/10.1128/AAC.48.7.2477-2482.2004

Cho, J., Choi, H., Lee, J., Kim, M.S., Sohn, H.Y. & Lee, D.G. (2013) The antifungal activity and membrane-disruptive action of dioscin extracted from *Dioscorea nipponica*. *Biochimica et Biophysica Acta*, 1828, 1153–1158. https://doi.org/10.1016/j.bbamem.2012.12.010

Clinical and Laboratory Standards Institute (2008) Reference method for broth dilution anti-fungal susceptibility testing of yeasts Approved standard-third edition. CLSI document M27-A3, 28, Wayne, PA.

Cutfield, S.M., Davies, G.J., Murshudov, G., Anderson, B.F., Moody, P.C., Sullivan, P.A. et al. (1999) The structure of the exo-beta-(1,3)-glucanase from *Candida albicans* in native and bound forms: relationship between a pocket and groove in family 5 glycosyl hydrolases. *Journal of Molecular Biology*, 294, 771–783. https://doi.org/10.1006/jmbi.1999.3287

Cortegiani, A., Misseri, G. & Chowdhary, A. (2019) What's new on emerging resistant *Candida* species. *Intensive Care Medicine*, 45, 512–515. https://doi.org/10.1007/s00134-018-5363-x

Davis, S.R. (2005) An overview of the *anti-Candida* properties of allicin and its breakdown products--the possibility of a safe and effective *anti-Candida* prophylactic. *Mycoses*, 48, 95–100. https://doi.org/10.1111/j.1439-0507.2004.01076.x

Deorukhkar, S.C., Saini, S. & Mathew, S. (2014a) (2014a) non-albicans *Candida* infection: an emerging threat. *Interdisciplinary Perspectives on Infectious Diseases*, 615, 958. https://doi.org/10.1155/2014/615958

Deorukhkar, S.C., Saini, S. & Mathew, S. (2014b) (2014b) virulence factors contributing to pathogenicity of *Candida tropicalis* and its *anti-Candida* susceptibility profile. *International Journal of Microbiology*, 456, 878. https://doi.org/10.1155/2014/456878

Fernandes, L., Oliveira, A., Henriques, M. & Rodrigues, M.E. (2020) Honey as a strategy to fight *Candida tropicalis* in mixed-biofilms with *Pseudomonas aeruginosa*. *Antibiotics (Basel)*, 9, 43. https://doi.org/10.3390/antibiotics9020043

Ferreira, L.G., Dos Santos, R.N., Oliva, G. & Andricopulo, A.D. (2015) Molecular docking and structure-based drug design strategies. *Molecules*, 20, 13,384–13,421. https://doi.org/10.3390/molecules200713384

Hayat, S., Cheng, Z., Ahmad, H., Ali, M., Chen, X. & Wang, M. (2016) Garlic, from remedy to stimulant: evaluation of *anti-Candida* potential reveals diversity in phytoalexin allicin content among garlic cultivars; allicin containing aqueous garlic extracts trigger antioxidants in cucumber. *Frontiers in Plant Science*, 7, 1235. https://doi.org/10.3389/fpls.2016.01235

Işcan, G., Kirimer, N., Kürkcüoğlu, M., Başer, K.H. & Demirci, F. (2002) Antimicrobial screening of *Mentha piperita* essential oils. *Journal of Agricultural and Food Chemistry*, 50, 3943–3946. https://doi.org/10.1021/jf011476k

Khodavandi, A., Harmal, N.S., Alizadeh, F., Scully, O.J., Sidik, S.M., Othman, F. et al. (2011) Comparison between allicin and fluconazole in *Candida albicans* biofilm inhibition and in suppression of HWP1 gene expression. *Phytomedicine*, 19, 56–63. https://doi.org/10.1016/j.phymed.2011.08.060

Koul, M., Meena, S., Kumar, A., Sharma, P.R., Singamaneni, V., Riyaz-Ul-Hassan, S. et al. (2016) Secondary metabolites from endophytic fungus *Penicillium pinophilum* induce ROS-mediated apoptosis through mitochondrial pathway in pancreatic cancer cells. *Planta Medica*, 82, 344–355.

Lanzotti, V., Scala, F. & Bonanomi, G. (2014) Compounds from *allium* species with cytotoxic and antimicrobial activity. *Phytochemistry Reviews*, 13, 769–791. https://doi.org/10.1007/s11101-014-9366-0

Le, A.V., Parks, S.E., Nguyen, M.H. & Roach, P.D. (2018) Improving the vanillin-Sulphuric acid method for quantifying Total saponins. *Technologies*, 6, 1-12.https://doi.org/10.3390/technologies6030084

Lemar, K.M., Turner, M.P. & Lloyd, D. (2002) Garlic (*Allium sativum*) as an anti-*Candida* agent: a comparison of the efficacy of fresh garlic and freeze-dried extracts. *Journal of Applied Microbiology*, 93, 398–405. https://doi.org/10.1046/j.1365-2672.2002.01707.x

Lewis, B.A. (1985) Inhibition of *Candida albicans* by methanethiol produced by *Brevibacterium linens*. *Microbiologica*, 8, 387–390 PMID: 3906369.

Lewis, J.S., 2nd & Graybill, J.R. (2008) Fungicidal versus fungistatic: what's in a word? *Expert Opinion on Pharmacotherapy*, 9, 927–935. https://doi.org/10.1517/14656566.9.6.927

Lima, S.L., Colombo, A.L. & de Almeida Junior, J.N. (2019) Fungal Cell Wall: emerging *anti-Candida*s and drug resistance. *Frontiers in Microbiology*, 10, 2573.

Mahajan, R. (2016) In vitro and cryopreservation techniques for conservation of Snow Mountain garlic. *Methods in Molecular Biology*, 1391, 335–346. https://doi.org/10.1007/978-1-4939-3332-7_23

Mangathayaru, K. (2013) *Pharmacognosy: an Indian perspective 1st ed. Pearson education in South Asia (chapter 6)*.India:Pearson Education India.

Martins, N., Barros, L., Henriques, M., Silva, S. and Ferreira, I.C. (2015) In vivo anti-*Candida* activity of phenolic extracts and compounds: future perspectives focusing on effective clinical interventions. *BioMed Research International* 2015, 247, 382. https://doi.org/10.1155/2015/247382.

Maubon, D., Garnaud, C., Calandra, T., Sanglard, D. & Cornet, M. (2014) Resistance of *Candida* spp. to *anti-Candida* drugs in the ICU: where are we now? *Intensive Care Medicine*, 40, 1241–1255. https://doi.org/10.1007/s00134-014-3404-7

Morita, T., Ushiroguchi, T., Hayashi, N., Matsuura, H., Itakura, Y. & Fuwa, T. (1988) Steroidal saponins from elephant garlic, bulbs of *Allium ampeloprasum* L. *Chemical & Pharmaceutical Bulletin (Tokyo)*, 36, 3480–3486. https://doi.org/10.1248/cpb.36.3480

Nakamoto, M., Kunimura, K., Suzuki, J.I. & Kodera, Y. (2020) Antimicrobial properties of hydrophobic compounds in garlic: allicin, vinyldithiin, ajoene and diallyl polysulfides. *Experimental and Therapeutic Medicine*, 19, 1550–1553. https://doi.org/10.3892/etm.2019.8388

Newman, D.J. & Cragg, G.M. (2012) Natural products as sources of new drugs over the 30 years from 1981 to 2010. *Journal of Natural Products*, 75, 311–335. https://doi.org/10.1021/np200906s

Perry, C.C., Weatherly, M., Beale, T. & Randriamahefa, A. (2009) Atomic force microscopy study of the antimicrobial activity of aqueous garlic versus ampicillin against *Escherichia coli* and *Staphylococcus aureus*. *Journal of the Science of Food and Agriculture*, 89, 958–964. https://doi.org/10.1002/jsfa.3538

Pettersen, E.F., Goddard, T.D., Huang, C.C., Couch, G.S., Greenblatt, D.M., Meng, E.C. et al. (2004) UCSF Chimera--a visualization system for exploratory research and analysis. *Journal of Computational Chemistry*, 25, 1605–1612. https://doi.org/10.1002/jcc.20084

Pristov, K.E. & Ghannoum, M.A. (2019) Resistance of *Candida* to azoles and echinocandins worldwide. *Clinical Microbiology and Infection*, 25, 792–798. https://doi.org/10.1016/j.cmi.2019.03.028

Said, M.M., Watson, C. & Grando, D. (2020) Garlic alters the expression of putative virulence factor genes SIR2 and ECE1 in vulvovaginal *C. albicans* isolates. *Scientific Reports*, 10, 3615. https://doi.org/10.1038/s41598-020-60,178-0

Sanguinetti, M., Posteraro, B. & Lass-Flörl, C. (2015) *Anti-Candida* drug resistance among *Candida* species: mechanisms and clinical impact. *Mycoses*, 58, 2–13. https://doi.org/10.1111/myc.12330

Seneviratne, C.J. & Rosa, E.A. (2016) Editorial: *anti-Candida* drug discovery: new theories and new therapies. *Frontiers in Microbiology*, 7, 728. https://doi.org/10.3389/fmicb.2016.00728

Shadkchan, Y., Shemesh, E., Mirelman, D., Miron, T., Rabinkov, A., Wilchek, M. et al. (2004) Efficacy of allicin, the reactive molecule of garlic, in inhibiting *aspergillus* spp. in vitro, and in a murine model of disseminated aspergillosis. *The Journal of Antimicrobial Chemotherapy*, 53, 832–836. https://doi.org/10.1093/jac/dkh174

Shah, N.C. (2014) Status of cultivated & wild *Allium* species in India: a review. *The Scitech Journal*, 1, 2347–7318.

Simons, V., Morrissey, J.P., Latijnhouwers, M., Csukai, M., Cleaver, A., Yarrow, C. et al. (2006) Dual effects of plant steroidal alkaloids on *Saccharomyces cerevisiae*. *Antimicrobial Agents and Chemotherapy*, 50, 2732–2740. https://doi.org/10.1128/AAC.00289-06

Sobolewska, D., Michalska, K., Podolak, I. & Grabowska, K. (2016) Steroidal saponins from the genus *allium*. *Phytochemistry Reviews*, 15, 1–35. https://doi.org/10.1007/s11101-014-9381-1

Sucher, A.J., Chahine, E.B. & Balcer, H.E. (2009) Echinocandins: the newest class of *anti-Candida*s. *The Annals of Pharmacotherapy*, 43, 1647–1657.

Taei, M., Chadeganipour, M. & Mohammadi, R. (2019) An alarming rise of non-*albicans Candida* species and uncommon yeasts in the clinical samples; a combination of various molecular techniques for identification of etiologic agents. *BMC Research Notes*, 12, 779. https://doi.org/10.1186/s13104-019-4811-1

Thamburan, S., Klaasen, J., Mabusela, W.T., Cannon, J.F., Folk, W. & Johnson, Q. (2006) *Tulbaghiaalliacea* phytotherapy: a potential anti-infective remedy for candidiasis. *Phytotherapy Research*, 20, 844–850. https://doi.org/10.1002/ptr.1945

Toral, L., Rodríguez, M., Martínez-Checa, F., Montaño, A., Cortés-Delgado, A., Smolinska, A. et al. (2021) Identification of volatile organic compounds in extremophilic bacteria and their effective use in biocontrol of postharvest fungal phytopathogens. *Frontiers in Microbiology*, 12, 773092. https://doi.org/10.3389/fmicb.2021.773092

Trease, G.E. & Evans, M.D. (1989) *A text book of pharmacognosy, 13thed*. London: Baillier, Tindal, & Caussel, pp. 144–148.

Walker, S.S., Xu, Y., Triantafyllou, I., Waldman, M.F., Mendrick, C., Brown, N. et al. (2011) Discovery of a novel class of orally active *anti-Candida* beta-1,3-D-glucan synthase inhibitors. *Antimicrobial Agents and Chemotherapy*, 55, 5099–5106. https://doi.org/10.1128/AAC.00432-11

Wang, Y., Wei, K., Han, X., Zhao, D., Zheng, Y., Chao, J. et al. (2019) The antifungal effect of garlic essential oil on *Phytophthora nicotianae* and the inhibitory component involved. *Biomolecules*, 9, 632. https://doi.org/10.3390/biom9100632

Yoshida, S., Kasuga, S., Hayashi, N., Ushiroguchi, T., Matsuura, H. & Nakagawa, S. (1987) *Anti-Candida* activity of ajoene derived from garlic. *Applied and Environmental Microbiology*, 53, 615–617. https://doi.org/10.1128/aem.53.3.615-617.1987

Zhu, Y.Z., Liu, J.W., Wang, X., Jeong, I.H., Ahn, Y.J. & Zhang, C.J. (2018) Anti-BACE1 and antimicrobial activities of steroidal compounds isolated from marine *Urechisunicinctus*. *Marine Drugs*, 16, 94. https://doi.org/10.3390/md16030094

Zúñiga-Martínez, M.L., Terán-Figueroa, Y., Vértiz-Hernández, A.A. & Alcántara-Quintana, L.E. (2019) Effect of Snow Mountain garlic extracts on cellular count and viability in cell lines cervical cancer. *International Journal of Ayurvedic and Herbal Medicine*, 9, 3596–3603. https://doi.org/10.31142/ijahm/v9i4.07

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